

Review



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Author for correspondence:

Sandra Rugonyi
e-mail: rugonyis@ohsu.edu

Quantifying blood flow dynamics during cardiac development: demystifying computational methods

Katherine Courchaine and Sandra Rugonyi

Biomedical Engineering, Oregon Health and Science University, Portland, OR 97239, USA

SR, 0000-0001-9262-7959

Blood flow conditions (haemodynamics) are crucial for proper cardiovascular development. Indeed, blood flow induces biomechanical adaptations and mechanotransduction signalling that influence cardiovascular growth and development during embryonic stages and beyond. Altered blood flow conditions are a hallmark of congenital heart disease, and disrupted blood flow at early embryonic stages is known to lead to congenital heart malformations. In spite of this, many of the mechanisms by which blood flow mechanics affect cardiovascular development remain unknown. This is due in part to the challenges involved in quantifying blood flow dynamics and the forces exerted by blood flow on developing cardiovascular tissues. Recent technologies, however, have allowed precise measurement of blood flow parameters and cardiovascular geometry even at early embryonic stages. Combined with computational fluid dynamics techniques, it is possible to quantify haemodynamic parameters and their changes over development, which is a crucial step in the quest for understanding the role of mechanical cues on heart and vascular formation. This study summarizes some fundamental aspects of modelling blood flow dynamics, with a focus on three-dimensional modelling techniques, and discusses relevant studies that are revealing the details of blood flow and their influence on cardiovascular development.

This article is part of the Theo Murphy meeting issue 'Mechanics of development'.

1. Introduction

Blood flow is essential for human life. In addition to the role blood plays delivering nutrients and oxygen to tissues, mechanical feedback from blood flow plays a fundamental role in the maintenance of a healthy cardiovascular system. Proper blood flow is required for cardiovascular health, and pathological conditions develop when flow is anomalous, particularly during embryonic stages of development. For the embryo, even small deviations from normal blood flow conditions can be far more devastating than for adults. Indeed, altered blood flow at early stages of embryonic development, before blood flow is necessary for oxygen and nutrient transport, is known to lead to congenital heart malformations [1–5]. This is because cells in the developing cardiovascular system sense and respond to mechanical cues, which influence growth and maturation of the cardiovascular system.

Embryonic development has always fascinated researchers. Not surprisingly, a wealth of information is available on how the embryo develops, and what genes and mechanisms are at play, for example [6,7]. Recent technological advances in molecular biology, including gene editing and high-throughput sequencing, are enabling researchers to determine the role of genes, and the consequences of genetic alterations, at an unprecedented level. However, when it comes to the essential role blood flow plays during development, much still needs to be explored. Among the challenges of understanding how

blood flow influences cardiovascular development are the daunting tasks of quantifying not only blood flow, but also the stresses exerted by blood flow on tissues. For biologists with no training in fluid mechanics, this can be particularly overwhelming. This review intends to demystify some of the aspects involved in computing blood flow dynamics, and to summarize what it takes to model blood flow in the developing cardiovascular system. Some examples are presented to show how computational fluid dynamics (CFD) can benefit research into mechanotransduction pathways that determine developmental responses to blood flow.

2. Embryonic cardiovascular development

Haemodynamics begins shaping the growth of the developing heart during the early embryonic stages. Blood circulation starts with the beating of the primitive tubular heart (around the beginning of gestational week 4 in human development) [8]. From this time on, the dynamics of blood flow dictates many aspects of cardiovascular development. For instance, genesis of new vessels and capillaries, and even differences between arterial and venous phenotypes, are determined by blood flow characteristics [9]. In the heart, the interaction between blood flow and cardiac tissues also determines how the heart continues to develop [8,10].

During tubular heart stages (approx. weeks 4–7 in human development), the pumping action of the heart creates a pulsatile-like blood pressure, and blood flows through the circulation. The combined effects of blood pressure and the frictional forces of flow on heart walls are sensed by cardiac tissues, and thus flow provides ‘feedback’ for proper cardiovascular development. Early on, for instance, development and regression of aortic arches, which will later give rise to the great arterial vessels, depend on blood flow conditions [11,12]; and in the heart outflow tract, increases in blood flow are followed by an increase in lumen diameter [13].

Endocardial cushions, which are protrusions of extracellular matrix (also known as cardiac jelly) towards the lumen, develop in the atrioventricular canal and outflow tract [14,15] of the tubular heart. Before valves develop, endocardial cushions act as primitive valves, and limit or prevent reverse flow [16]. Cushions are populated by mesenchymal cells prior to their development into mature valves. This is accomplished by an endocardial–mesenchymal transition (EMT), in which endocardial cells acquire a mesenchymal phenotype and start migrating towards the cardiac jelly [17]. Several studies have established the importance of blood flow during valve development, including early EMT stages [18–22]. Altered blood flow leads to improper EMT and, later on in development, to valve and septal defects [2,23,24].

Meanwhile, by the end of week 4 in human gestation, the primitive ventricle starts developing trabecular structures, which resemble a sponge-like porous structure [25,26]. These trabecular structures aid ventricular contraction, and their architecture is influenced by blood flow conditions [26–29]. Indeed, increased blood pressure in the ventricle during development leads to thickening of both the compact myocardium and trabeculae [26]. Defects in the early trabecular architecture can later lead to cardiac defects including hyper- or hypo-trabeculation, which are both associated with cardiomyopathies [30].

The fully formed, four-chambered heart continues to be influenced by blood flow conditions, both before and after birth. During fetal stages, for example, increased blood pressure leads to thicker cardiac walls, as in the adult heart [31–34]. However, little is known about other early remodelling effects induced by anomalous cardiac blood flow during embryonic and fetal stages. Understanding these effects is important for the treatment of babies with congenital heart defects. The presence of a defect necessarily leads to anomalous blood flow in the heart, but the repercussion of these flow anomalies on cardiac tissues is not known.

During embryonic developmental stages and beyond, blood flow provides mechanical feedback that regulates cardiovascular development and maintenance together with genetic programmes. In the adult cardiovascular system, many of the detrimental effects of anomalous blood flow are understood and can even be modelled mathematically [35–38]. During developmental stages, however, more research is needed to understand how exactly blood flow dynamics affects cardiac development. Embarking upon such studies requires quantification and analysis of blood flow forces acting on cardiovascular tissues, a task that is often accomplished through computational modelling.

3. Cardiovascular modelling

Mathematical models of cardiovascular flow dynamics vary in their level of complexity and in the distinctive details their solutions provide. The simplest model is the so-called Windkessel-type lumped parameter model of flow, also known as a zero-dimensional model [39]. Despite their simplicity, Windkessel models can represent the dynamics of the entire circulatory system. They are analogous to an electrical circuit, with segments and elements representing different portions and properties of the circulation: resistance elements represent resistance to flow by viscosity; capacitance-like elements represent the compliance of cardiovascular walls; and inductance-like elements represent flow inertia effects. The heart itself can be modelled as a pressure/flow source, with different properties and functions to reproduce Starling’s Law of the heart [40]. In Windkessel-type models the two main variables being solved for are the flow rate (Q) and blood pressure (P), and solutions to the model predict how these variables change over time and in different parts of the circulation.

Another frequently used model is the one-dimensional pressure wave propagation model [41], which describes how an increase in blood pressure travels along a vessel and deforms its walls. The wave propagation model includes small deformations of the arterial wall as the pressure wave travels through it, and thus the vessel area is solved for in addition to Q and P . Because changes in arterial diameter can be imaged in the large arteries of the body, this model can be used in conjunction with *in vivo* imaging data to estimate arterial wall properties. To account for the connection of the artery to the rest of the circulation, zero-dimensional (Windkessel) models are often used together with wave propagation models [42,43].

Finally, three-dimensional models of portions of the vasculature or the heart (and two-dimensional models as special cases) are used to solve for the details of blood flow velocities. These include vortexes, recirculation regions and the gradient of velocities near cardiovascular walls (from which wall shear

rates and stresses are derived). The vascular geometry used in these models can be constructed to a theoretical ideal (e.g. a straight circular cylinder) or extracted from patient (or animal) images, enabling 'subject-specific' simulations. These complex three-dimensional models are typically the most involved, and are described in more detail in the following sections.

4. Modelling blood flow and the Navier–Stokes equations

When modelling the dynamic flow of blood in the embryonic cardiovascular system, it is first necessary to consider and account for the rheological properties of the blood. Plasma is a Newtonian fluid, but the addition of blood cells and other molecules in plasma renders blood non-Newtonian [44,45]. To understand the difference between Newtonian and non-Newtonian fluids, let us consider a simple case of flow between parallel plates, which are separated by a small distance h (figure 1). The lower plate is fixed in space, immobile; while the upper plate is pushed with a force F , which results in the plate moving at a constant velocity V . Owing to the contact with the plates, fluid velocity is zero at the lower plate ($y = 0$) and V at the upper plate ($y = h$). If the fluid between the plates is Newtonian, then the relationship between F and V is linear, and the velocity profile in between the plates is also linear ($v_x = V/h \ y$). This means that the change in velocity relative to the change in distance from the plate, also called shear rate ($\gamma = dv_x/dy = V/h$), is constant. The shear stress (τ) for this flow is simply the force F per unit area A (this is the area of the upper plate in contact with the fluid). For a Newtonian fluid, the relationship between τ and γ is linear and the proportionality constant is the fluid's viscosity coefficient, μ . In a non-Newtonian fluid, the relationship between τ and γ is no longer linear, and therefore μ is no longer a constant but depends on flow variables such as shear rate, i.e. $\mu = \mu(\gamma)$. In particular, blood viscosity decreases as the shear rate increases, a behaviour also known as shear thinning [46]. Non-Newtonian models frequently employed to describe the behaviour of blood flow include the Carreau–Yasuda shear thinning model [47], as well as Lattice Boltzmann models, which explicitly model particle suspensions (red blood cells in plasma) [48]. Modelling the behaviour of non-Newtonian fluids is typically much more involved than modelling a Newtonian fluid. There are, however, some cases in which blood flow can be approximated as a Newtonian flow. Two cases are of importance: (i) when shear rates are high; (ii) when the haematocrit (Ht) is low. For normal Ht levels (45%), the relationship between shear stress and shear rates becomes linear at high shear rates (greater than 100 s^{-1}) and the flow can be assumed to be Newtonian [44]. This assumption is frequently used to model flow in the mature heart and large arteries. When the volume of red blood cells in blood is low (low Ht), as it is at early stages of embryonic cardiovascular development (Ht < 20%) [45], the behaviour of blood flow is similar to that of plasma, and blood can be assumed to be Newtonian in the heart and larger vessels (although not in capillaries). This assumption has been used to model blood flow in the heart and aortic arches of early embryos. It is also important to note that blood viscosity depends on Ht [45,46]: viscosity increases with increased Ht. As such, even when blood is

modelled as Newtonian, changes in Ht with embryonic stage need to be accounted for with changes in μ .

To model the flow of blood, equations of motion for the fluid need to be established. These equations are basically derived under the assumption that every portion of the fluid (each infinitesimal volume, which we will call a 'particle') satisfies the laws of physics, and in particular Newton's Law of Motion, which establishes that the mass of a particle multiplied by its acceleration is equal to the sum of forces applied to the particle. Mathematically

$$m \frac{D\mathbf{v}}{Dt} = \sum_i \mathbf{F}_i, \quad (4.1)$$

where m is the mass, the acceleration is the derivative of the velocity vector $\mathbf{v}$ with respect to time and forces are represented by the vectors $\mathbf{F}_i$. In equation (4.1), $D\mathbf{v}/Dt$ represents the 'material derivative' of velocity: that is, the change of the particle velocity with respect to time as we follow the moving particle. These equations are written in the Lagrangian formulation of motion, which is based on the perspective of an observer who follows the particles as they move. The Lagrangian formulation is typically used to describe the motion of solids.

(a) Constitutive equations

In a fluid, forces applied to each 'particle' come from the fluid pressure and frictional forces among nearby particles moving at different velocities. To represent these forces in the flow, stresses are used. The simplest way to think of stresses is as forces per unit surface area. Thus, for each given surface there are three stress components (figure 2), representing forces perpendicular to the surface (normal stresses) and forces tangential to the surface (shear stresses). In mathematical terms, stresses are represented by an entity called the stress 'tensor' ($\boldsymbol{\sigma}$), which is usually written as a matrix:

$$\boldsymbol{\sigma} = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{pmatrix}, \quad (4.2)$$

where σ_{xx} , σ_{yy} and σ_{zz} are the components of the normal stresses (in the x , y and z directions, respectively); σ_{xy} , σ_{yx} , σ_{xz} , σ_{zx} , σ_{yz} and σ_{zy} are the components of shear stresses. Moreover, $\sigma_{xy} = \sigma_{yx}$, $\sigma_{xz} = \sigma_{zx}$ and $\sigma_{yz} = \sigma_{zy}$. The stress tensor of a Newtonian fluid can be represented by its components, as follows:

$$\left. \begin{aligned} \sigma_{xx} &= -p + 2\mu \left(\frac{\partial v_x}{\partial x} \right), \\ \sigma_{yy} &= -p + 2\mu \left(\frac{\partial v_y}{\partial y} \right), \\ \sigma_{zz} &= -p + 2\mu \left(\frac{\partial v_z}{\partial z} \right), \\ \sigma_{xy} &= \sigma_{yx} = \mu \left(\frac{\partial v_y}{\partial x} + \frac{\partial v_x}{\partial y} \right), \\ \sigma_{xz} &= \sigma_{zx} = \mu \left(\frac{\partial v_z}{\partial x} + \frac{\partial v_x}{\partial z} \right) \\ \text{and } \sigma_{yz} &= \sigma_{zy} = \mu \left(\frac{\partial v_z}{\partial y} + \frac{\partial v_y}{\partial z} \right), \end{aligned} \right\} \quad (4.3)$$

where p is the flow pressure; x , y and z are coordinates; v_x , v_y and v_z are the components of the flow velocity vector, $\mathbf{v}$, in the x , y and z directions, respectively; and μ is the viscosity

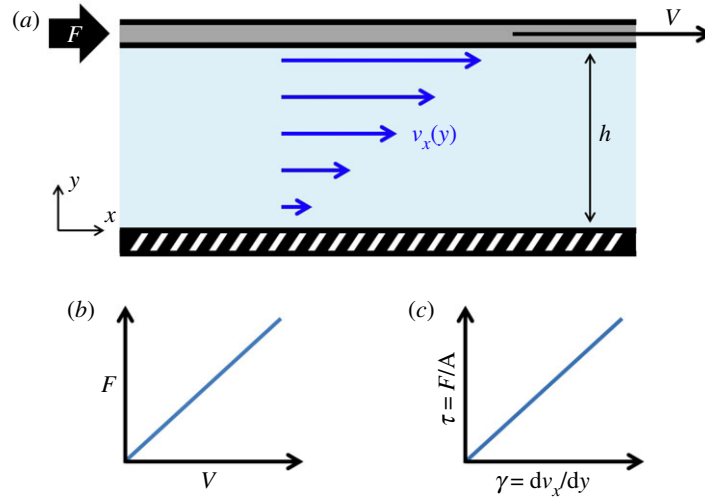


Figure 1. Newtonian flow between parallel plates. (a) The lower plate is fixed in space, while the upper plate is pushed with a force F , which results in the plate moving at a constant velocity V . The velocity profile of the fluid in between the plates is linear ($v_x = V/h y$). (b) The relationship between F and V is linear for a Newtonian fluid. (c) The relationship between τ , the shear stress (force F per unit area A of the upper plate that is in contact with the fluid), and γ , the shear rate (constant, $\gamma = dv_x/dy = V/h$), is linear, and the proportionality constant is the fluid's viscosity, μ .

coefficient. Note that the flow between parallel plates discussed before is a special case, in which $\sigma_{xy} = \sigma_{yx} = \mu (\partial v_x / \partial y)$, and all other stress components are zero.

A tensor allows a single mathematical entity to represent the forces acting on any surface of a particle. To get the stresses applied to a surface, also called traction forces, $\mathbf{t}$, we first need to compute the unit vector, $\mathbf{n}$, perpendicular to the surface of interest. Then the traction force is simply

$$\mathbf{t} = \boldsymbol{\sigma} \cdot \mathbf{n}. \quad (4.4)$$

When using Cartesian coordinates (x, y, z) , equation (4.4) simply involves multiplying a matrix by a vector.

Combining Newtonian fluidic behaviour (equation (4.3)) with Newton's Laws of Motion for each particle (equation (4.1)) results in the well-known Navier–Stokes equations for modelling flow. When gravitational effects are neglected, which is frequently appropriate during developmental stages, the Navier–Stokes equations are

$$\left. \begin{aligned} \rho \left(\frac{\partial v_x}{\partial t} + v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} + v_z \frac{\partial v_x}{\partial z} \right) &= -\frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} + \frac{\partial^2 v_x}{\partial z^2} \right), \\ \rho \left(\frac{\partial v_y}{\partial t} + v_x \frac{\partial v_y}{\partial x} + v_y \frac{\partial v_y}{\partial y} + v_z \frac{\partial v_y}{\partial z} \right) &= -\frac{\partial p}{\partial y} + \mu \left(\frac{\partial^2 v_y}{\partial x^2} + \frac{\partial^2 v_y}{\partial y^2} + \frac{\partial^2 v_y}{\partial z^2} \right), \\ \text{and } \rho \left(\frac{\partial v_z}{\partial t} + v_x \frac{\partial v_z}{\partial x} + v_y \frac{\partial v_z}{\partial y} + v_z \frac{\partial v_z}{\partial z} \right) &= -\frac{\partial p}{\partial z} + \mu \left(\frac{\partial^2 v_z}{\partial x^2} + \frac{\partial^2 v_z}{\partial y^2} + \frac{\partial^2 v_z}{\partial z^2} \right), \end{aligned} \right\} \quad (4.5)$$

where ρ is the fluid density, p is the hydrostatic pressure and μ is the viscosity coefficient.

In vector notation, which is more compact, the Navier–Stokes equations become

$$\rho \left[\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} \right] = -\nabla p + \mu \nabla^2 \mathbf{v}. \quad (4.6)$$

Unlike equation 4.1, equations 4.5 and 4.6 express equilibrium of forces per unit volume (thus the use of density, ρ , instead of mass, m), and are written in the Eulerian formulation of motion, which describes motion from the

perspective of an observer who is not following the particles as they move but rather is interested in describing how velocities change at a specific region in space (e.g. a portion of a blood vessel, or a specific point inside the vessel lumen). The term between brackets on the left-hand side of equation (4.6) represents the material derivative $D\mathbf{v}/Dt$ of equation (4.1) written in the Eulerian formulation: $\partial \mathbf{v} / \partial t$ is the change in velocity at a specific location (point) in the fluid (which is not equal to the change in velocity of a particle, as many particles pass through the point over time), and $(\mathbf{v} \cdot \nabla) \mathbf{v}$ represents flow convection, which accounts for transport within the fluid domain (as particles pass through a particular point, they carry with them their inertia and properties to neighbouring locations). The right-hand side represents the effect of pressure (first term) and frictional forces due to flow motion (second term).

Equations (4.5) and (4.6) express that flow is the result of pressure gradients (∇p), which are counteracted by viscous

forces ($\mu \nabla^2 \mathbf{v}$); fluid inertia and convection (left-hand term), which account for flow transport of particles, also play a role in balancing forces.

(b) Boundary conditions

Because these equations are partial differential equations, boundary conditions are required to solve them. Boundary conditions specify how the fluid enters and leaves the portion of the vasculature that we are modelling. The boundaries of our problem are the surface of the vessel wall in contact

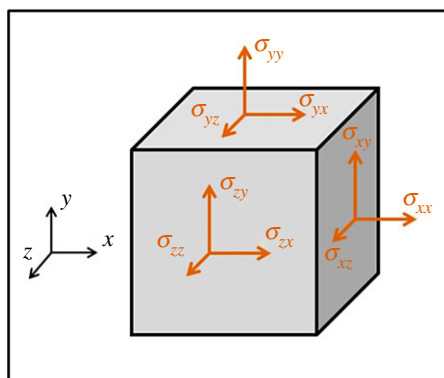


Figure 2. Stresses on a surface. For every infinitesimal surface in a flow, there are stresses normal to the surface (σ_{xx} , σ_{yy} and σ_{zz}) and stresses tangential to the surface (σ_{xy} , σ_{yx} , σ_{xz} , σ_{zx} , σ_{yz} and σ_{zy}). In a tissue, blood pressure imposes stresses normal to the tissue wall and the flow of blood imposes wall shear stresses tangential to the wall surface, at the blood–tissue contact interface. These stresses are calculated as $\boldsymbol{\sigma} \cdot \mathbf{n}$, where $\mathbf{n}$ is the normal to the surface, and can be visualized from this figure as a change in coordinate orientation that aligns the surface of interest with the tissue surface.

with the fluid and the inlets and outlets of the portion of the vasculature considered. We represent these boundary surfaces by Γ . Then, we have two possible boundary conditions that we can apply to a boundary surface: either a traction force boundary condition (force per unit area, e.g. pressure), or a velocity boundary condition. For example, if we know the rate at which blood is entering the heart or vessel segment, we can impose it as a velocity boundary condition (typically as either a flat, constant velocity profile or parabolic velocity profile). If instead we know the pressure drop in a vascular segment, we can impose an inlet and outlet traction force (approx. pressure). The boundary surface(s) are divided into a portion in which traction forces are applied, Γ_t , and a portion in which velocities are applied, Γ_v . Mathematically, the boundary conditions are expressed as

$$\begin{aligned} \boldsymbol{\sigma} \cdot \mathbf{n} &= \mathbf{t}^s & \text{in } \Gamma_t \\ \mathbf{v} &= \mathbf{v}^s(\mathbf{x}, t) & \text{in } \Gamma_v, \end{aligned} \quad (4.7)$$

where $\mathbf{t}^s$ is the traction force applied to the boundary surface and $\mathbf{v}^s$ is the flow velocity applied to the boundary surface.

For the mathematical problem to be well defined, we cannot apply both traction and velocity to the same boundary surface, and every boundary surface has to have a boundary condition. Mathematically,

$$\begin{aligned} \Gamma &= \Gamma_t \cup \Gamma_v \\ 0 &= \Gamma_t \cap \Gamma_v. \end{aligned} \quad (4.8)$$

The Navier–Stokes equations (equations (4.5) and (4.6)) can be solved analytically in a number of simple cases. Perhaps the most relevant to circulation flows is the solution of Poiseuille flow, or flow in a cylindrical tube. Typical boundary conditions are zero velocity at the cylindrical tube walls (also known as the no-slip boundary condition), and a uniform pressure at the inlet and a different uniform pressure at the outlet of the tube (which together impose a pressure gradient in the tube). The solution to this problem can be found in any book on fluid mechanics, and it establishes that the velocity profile inside the tube is parabolic and depends on the pressure gradient.

When modelling blood flow through a cardiovascular segment, typically the outlet boundary conditions are not known. This is because flow depends not only on the geometry and properties of the segment we are modelling, but also on the rest of the circulatory system to which the segment is connected. As such, while we can measure or approximate the inlet conditions for the segment, the outlet conditions (e.g. pressure, flow splits on bifurcations) are not obvious. To circumvent these limitations, and correctly capture the physiology of blood flow in the segment modelled, zero-dimensional ‘Windkessel-type’ boundary conditions are imposed [49–52]. Basically, we couple the solution of the three-dimensional segment with the solution of a zero-dimensional model of the circulation.

5. Computational models of the developing heart and vascular vessels

When solving more involved problems, such as solution of flows in segments of the vasculature that are not straight, vessel bifurcations or vessel enlargements, analytical solutions do not exist. Semi-analytical solutions or simplifications of the models are sometimes used, but in other cases investigators resort to computational or numerical techniques to solve the problem. Owing to the relatively easy access to powerful computers, the use of CFD is becoming the ‘gold standard’ in many fields, including those that seek to quantify flow mechanics during heart development. The main advantage of computational modelling and simulation is that once flow velocities are computed, stresses, including wall shear stresses (which are the stresses tangential to the cardiovascular wall due to friction of flowing blood), and pressures can be quickly and simply extracted from the results, and even plotted in four dimensions (three dimensions over time).

Computational modelling of flow dynamics involves discretization of the equations of motion (equation (4.6)). Discretization is typically performed using a grid or mesh that basically divides the geometrical volume of the model into smaller parts. This is needed to express the equations of motion in discrete, algebraic equations at each grid point, so that the computer can solve a system of algebraic equations rather than continuous equations at the infinite points that form a volume. The discretization of equations is not arbitrary and follows special rules, depending on what technique is employed (e.g. finite-element methods, finite-volume techniques). While discretization techniques employed (and the mathematics behind them) are outside the scope of this review, it is worth noting that the discretization chosen has to be such that numerical instabilities are avoided, and numerical errors are minimized. Numerical instabilities and errors cause results to deviate, sometimes quite significantly, from the actual solution. To avoid this, a careful choice of discretization techniques and number of discretized elements is needed. The interested reader is referred to textbooks and works on the subject, e.g. [53,54].

A major consideration when designing a model of blood flow is whether to include blood–tissue interactions (figure 3). Blood in major cardiovascular segments (i.e. not capillaries and blood islands) is contained inside arterial, venous and cardiac walls, and flow therefore interacts with cardiovascular tissues. While it is theoretically possible to

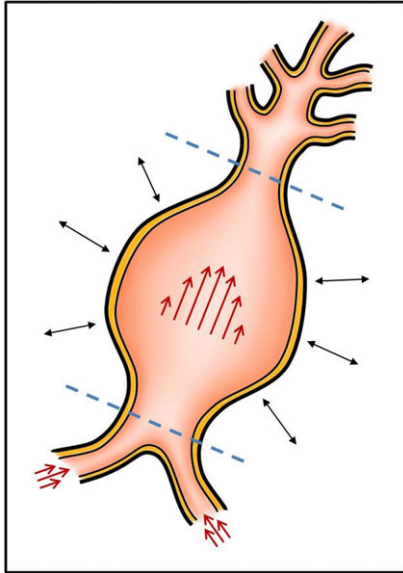


Figure 3. Schematics of the developing tubular heart. The portions of the vasculature above and below the dashed lines (outside the heart) are typically modelled using a fixed geometry, while the heart itself is typically modelled using moving-wall models.

fully model this blood–tissue interaction, tissue properties are frequently unknown and the modelling is computationally more expensive (memory, amount of computations performed, time per simulation). In many cases, it is not necessary to fully model the blood–tissue interaction, and simplifications are made. In these cases, the solution of blood flow dynamics is separated from the solution of tissue mechanics. The simplest possible case is when we can neglect the expansion and contraction of cardiovascular walls during the cardiac cycle. In this case, the geometry of the vessel or vessels of interest can be extracted or idealized from a single static image. We can still simulate pulsatile flows, in which flow dynamics changes over time, but without inducing wall motion. When the motion of cardiovascular walls cannot be neglected, as in the case of the beating heart, we can incorporate this motion into the model. Cardiovascular wall motion is either idealized or extracted from dynamic images. The different scenarios in modelling blood flow (fixed walls, moving walls and explicit blood–tissue interaction) are considered in what follows.

(a) Modelling blood flow in cardiovascular segments

When modelling blood flow outside the heart, such as in aortic arches, it is frequently assumed that vessel walls do not move. The small-amplitude motion of the vessel (expansion/contraction) is neglected, and a no-slip condition (zero velocity) is assumed at the vessel walls. The geometry of the portion of the vasculature to be modelled is then generated and discretized. This geometry can be idealized, or extracted (segmented) from images (e.g. micro-CT, confocal). Imaged-based geometries allow subject-specific modelling, which can be used to compare individual flow dynamics within groups and determine differences between groups (e.g. treated versus control). Navier–Stokes equations (equation (4.6)) can then be used to model the flow of blood within the region of interest. Boundary conditions at the inlet(s) and outlet(s) of the model are then imposed to further specify how blood is flowing through the vessel(s).

These boundary conditions basically establish either a pressure gradient in the blood vessel(s) or the volume flow rate through the vessel(s). CFD models of the developing cardiovascular system have been developed with the objective of assessing blood flow dynamics in the developing circulation and its effects on further development.

Fixed-wall CFD models of the developing aortic arches have been used to determine how blood flow conditions influence the generation and regression patterns of the arches [11,12]. Aortic arches, which are vessels branching from the aortic sac in pairs, are naturally generated and regressed in a predefined order over developmental time, leading to dynamic aortic arch patterns. These patterns depend on haemodynamic conditions and can be altered when blood flow differs from normal, eventually leading to aberrant morphologies of the great vessels [11]. In a recent study, arch geometries were obtained from micro-CT images of the chicken embryo vasculature after polymeric casting. CFD models were then used to quantify flow patterns through the arches during tubular heart stages in chicken embryos (figure 4a). These models, together with experimental measurements of arch diameters, reveal that wall shear stress in the arches is correlated with arch diameter [12] and that the orientation of the outflow tract portion of the heart (which is just upstream of the arches) influences blood flow distributions and dynamics within the arches [11]. CFD models in combination with experimental data revealed the importance of blood flow on aortic arch patterning, and could be used to predict pattern formation, gaining new insights into how blood flow conditions influence cardiovascular growth and morphogenesis [4].

(b) Modelling moving walls to simulate cardiac pumping

Perhaps the simplest way to model the dynamics of blood flow within the pumping heart is by imposing a cyclic motion on the fluid domain boundaries that simulates the motion of cardiac walls (figure 3). If this is the case, we do not need to know the contractile or passive properties of cardiac walls, and instead we assume that we know the wall motion resulting from the blood–tissue interaction. Accurate cardiac wall motion can be extracted from dynamic images of the beating heart. In this case, we segment the position of cardiac walls over time and impose wall motion onto our models.

The Navier–Stokes equations are then slightly modified to account for the changing flow domain due to the moving walls:

$$\rho \left\{ \frac{\partial \mathbf{v}}{\partial t} + [(\mathbf{v} - \hat{\mathbf{v}}) \cdot \nabla] \mathbf{v} \right\} = -\nabla p + \mu \nabla^2 \mathbf{v}, \quad (5.1)$$

where $\hat{\mathbf{v}}$ is the velocity at which the grid or mesh is moving as the cardiac fluid volume changes over time. At the boundaries representing the cardiac walls, $\hat{\mathbf{v}}$ is just the velocity of the wall, $\mathbf{v} = \hat{\mathbf{v}}$: that is, the fluid in contact with the wall moves together with the wall (this is the no-slip boundary condition). Inside the domain, $\hat{\mathbf{v}}$ accounts for differences between the motion of fluid particles and the motion of the grid/mesh representing the fluid domain. If the mesh is not moving, $\hat{\mathbf{v}} = 0$, we recover the Navier–Stokes equations for a fixed domain, equation (4.6).

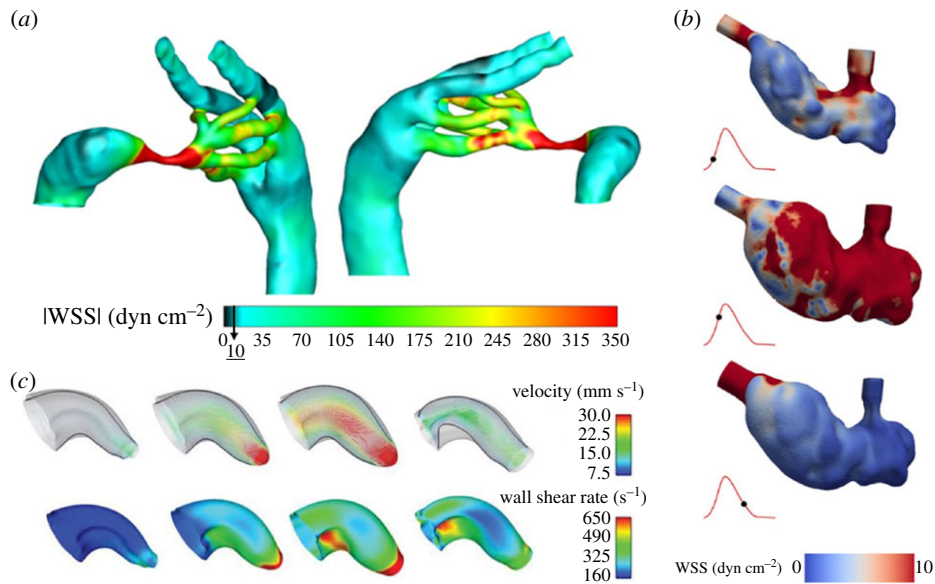


Figure 4. Applications of CFD modelling to embryonic development. (a) Wall shear stress in one configuration of the developing aortic arches of chicken embryos at Hamburger–Hamilton stage HH21 (tubular heart stage), as viewed from the left and right sides. Reproduced with permission from Kowalski *et al.* [12]. (b) Model of wall shear stress profiles over the cardiac cycle in the primitive ventricle of zebrafish during trabeculation stages. Reproduced with permission from Vedula *et al.* [55]. (c) CFD modelling of the HH18 embryonic heart outflow tract over the cardiac cycle. Reproduced with permission from Midgett *et al.* [13].

Equation (5.1) is written in what is known as an arbitrary Lagrangian–Eulerian (ALE) formulation of motion [56]. In a Lagrangian formulation of motion (typically used to describe the motion of solid structures and tissues), particles are followed in their motion and the grid/mesh moves and deforms following the particles ($\mathbf{v} = \hat{\mathbf{v}}$). In an Eulerian formulation of motion (typically used to describe flows), the volume in which the fluid flows is fixed in space and we describe the motion of particles flowing through the volume (e.g. a pipe or blood vessel) but we do not follow the motion of the particles after they leave the domain or before entering it ($\hat{\mathbf{v}} = 0$). An ALE formulation is basically a combination of both a Lagrangian and a Eulerian formulation of motion. Indeed, when $\hat{\mathbf{v}} = 0$, we recover the Eulerian formulation of equation (4.6); when $\mathbf{v} = \hat{\mathbf{v}}$, we recover the Lagrangian formulation, equation (4.1), in which the grid follows the motion of the particles.

From a computational and mathematical modelling point of view, imposing the motion of cardiac walls to simulate heart pumping is simpler than simulating the interaction of cardiac tissue and blood flow. From a more physical perspective, however, it is difficult to know *a priori* how the heart walls will move, especially under abnormal conditions, such as when altering cardiac or flow parameters. To circumvent this difficulty, image-based heart wall motion can be extracted from embryo-specific four-dimensional cardiac images. Of course this requires acquiring good images of the developing heart, and accurately segmenting those images (extracting the boundaries of interest) over space and time. This approach has been employed with zebrafish and chicken embryonic heart models, as well as human fetal heart models. Image-based simulation of cardiac pumping enables embryo-specific simulation of cardiac flow that retains the complex physiological motion of cardiac walls.

CFD models of the zebrafish embryonic ventricle have been developed to assess the biomechanical effects of flow due to different interventions and mutations [57]. The zebrafish embryo is frequently used for studying cardiac

development because of its tissue transparency, ease of genetic manipulations and easy access for imaging. Transgenic lines with fluorescently labelled cells have been used to image cardiac motion with light sheet microscopy techniques [58], with enough spatial and temporal resolution to resolve the motion and geometry of the heart. During imaging, the motion of several cardiac planes is acquired and then algorithms are used to reconstruct the heart motion in four dimensions (three dimensions over time). Segmentation of these four-dimensional images provides the cardiac wall motion for use in CFD models of the beating heart (figure 4b). Image-based CFD models have been used to model ventricular haemodynamic conditions in zebrafish under normal conditions, and under the effects of mutations that alter blood flow viscosity and cardiac contraction [55]. These studies are being used to determine the effect of haemodynamic conditions on ventricular trabeculation and growth.

Image-based CFD models of the chick embryonic outflow tract during tubular stages of heart development have also been developed and successfully implemented [59,60]. The chicken embryo is frequently employed in the study of heart development because of its transparency and ease of access for manipulation and imaging. Although transgenic lines are uncommon, imaging can be performed by optical coherence tomography (OCT), which enables acquisition of structural images and Doppler velocities with enough spatial and temporal resolution to resolve the cardiac motion and geometry of the heart. As in the zebrafish example, images along a series of cardiac planes are acquired and then algorithms are used to reconstruct the four-dimensional motion of the heart outflow tract. CFD models of the outflow tract (figure 4c) have revealed that wall shear stresses are increased in endocardial cushions, which will develop into semilunar valves and a portion of the interventricular septum. Interventions that alter haemodynamic conditions in the chick heart influence these wall shear stresses in the cushions, leading to conotruncal heart defects [2]. CFD models, together with

experimental data, are being used to study how mechano-transduction mechanisms affect valve formation and cardiac septation.

Lastly, image-based CFD models of the human fetal heart are being developed to understand the role of haemodynamics in normal and malformed hearts [61,62]. The models so far are based on four-dimensional ultrasound scans of 20-week-old normal human fetuses. Image analyses of ultrasound scans revealed the motion of the heart walls that was used in the generation of CFD models of the fetuses' left and right ventricles. In both cases (left and right ventricles), simulations revealed complex flow patterns and the presence of flow vortex rings, which generate significant wall shear stress on the endocardium and may play an important role on cardiac efficiency [61].

(c) Modelling the interaction of blood flow and cardiac tissue mechanics

If cardiac tissue properties are known (including passive and active contractile properties), then we can model the cyclic contraction and expansion of the heart, and its interaction with blood flow dynamics. This is a multi-physics problem that involves simultaneous solution of blood flow dynamics within the heart and cardiac tissue motion and deformation. This problem is referred to (in the engineering community) as a fluid–structure interaction (FSI) problem, as it involves solution of a fluid flow interacting with a solid 'structure' (the cardiac tissue). An accurate FSI model of the developing heart would be ideal to investigate the response of the heart to changes in flow, pressure and other haemodynamic or tissue variables. However, this model is the hardest to achieve, and has not been attempted in the modelling of the developing heart,

mainly because developing cardiac tissue properties are difficult to measure, and thus model accurately. This is, however, the logical next step in modelling cardiac mechanics during development.

6. Conclusion

Modelling blood flow dynamics during cardiac development is a promising area for further research and development. Detailed CFD models of both the developing vasculature and the heart, together with experimental data on adaptations to blood flow, are starting to elucidate important aspects of the intricate mechanisms by which blood flow dynamics regulates cardiovascular growth and development. Wall shear stresses play a substantial role in cardiovascular adaptations to flow, but they are difficult to estimate using only measured flow data. Once blood flow dynamics have been simulated, CFD models offer the advantage of relatively easy quantification of wall shear stresses on endothelial and endocardial walls, derived from flow variables. Research integrating CFD modelling and molecular biology techniques will certainly continue to decipher how blood flow dynamics influences cardiac formation and malformation.

Data accessibility. This article has no additional data.

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